

PROFESSIONAL NURSING COACH

PHARMACOLOGY & GENETICS

PNC ULTRA-LMR HYBRID MASTER COMPENDIUM • NUR-09 • VERIFIED

Pharma Units I–VIII + Genetics • 6 batches • 94 LMR pages + cover & TOC • NORCET / RRB / ESIC

Batch	Units	Pp
P-B1	Pharma Unit I	18
P-B2	Pharma Units II-IV	18
P-B3	Pharma Unit V-A	16
P-B4	Pharma Units V-VII	15
P-B5	Pharma Unit VIII	15
G-B1	Genetics (Hybrid-C)	12

WhatsApp DM: +91 90571 22020 • v1.0 Sep-2026 • Pharma Classic + Genetics Hybrid-C (Pathology has its own Verified file)

Pp	Block	Focus
4-21	PHAR-B1	Pharma Unit I
22-39	PHAR-B2	Pharma Units II-IV
40-55	PHAR-B3	Pharma Unit V-A
56-70	PHAR-B4	Pharma Units V-VII
71-85	PHAR-B5	Pharma Unit VIII
86-97	GEN-B1	Genetics: inheritance + disorders + counseling

How to use: revise batch-wise with mocks; log errors. Double-QA pending sign-off.

Q	Standard	Status
Q1	All batches merged in unit order	Done
Q2	Classic batches preserved; new batches Hybrid-C	Done
Q3	Footers + TOC ranges verified	Done

DEFINITIONS, BRANCHES & SOURCES

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A+

V2 DENSE

Core: **Pharmacology** = drug study; **pharmacy** = dispensing; toxicology = poisons. Sources decide allergy + counsel.

Pharmacokinetics	What body does to drug (ADME) — dose/frequency/liver/kidney
Pharmacodynamics	What drug does to body (receptor) — effect + adverse
Chemotherapy	Selective kill pathogen/cancer; resistance + stewardship
Plant source	Atropine-belladonna, digoxin-foxglove, morphine-poppy
Animal/mineral	Insulin, heparin, iron, calcium — allergy + diet link
Biotech/synthetic	Recombinant insulin, mAbs — cold chain + cost

- Branches: pharma-study, pharmacy-dispense, toxicology-poison, posology-doses.
- Chemotherapy = selective toxicity; narrow TI needs levels.
- Ask source allergy (penicillin-mould, egg-vaccine) + diet + herbal.

★ **Clinical Pearl:** Bedside pearl: every drug history asks **allergy + pregnancy + liver/kidney + OTC/herbal** before first dose.🔪 **NORCET Drill:** Mini-MCQ: Body handles drug is? (a) PD (b) **PK/ADME** (c) Toxicology — Ans (b).**FATAL TRAP:** Pharmacy = pharmacology. Study vs dispense differ.**NORCET HIT:** Body-drug vs drug-body.

DOSAGE FORMS & ROUTES MASTER

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V2 DENSE

Core: **Enteral slow-cheap**, **parenteral fast-sterile**, topical/inhaled local. Choice = speed + site + cooperation + first-pass.

Oral	Slow convenient; first-pass cuts morphine/propranolol; food/pH matters
IV	Immediate + titratable ; emergency, shock, fluids; infection/phlebitis watch
IM/SC	Moderate; vaccine/insulin/heparin; shock-edema poor uptake
Sublingual/buccal	Fast bypass first-pass ; nitro/buprenorphine; dry mouth fail
Inhaled/topical/rectal	Airway/skin/child-vomit; technique + barrier + consent
Enteric/sustained	Never crush ; plain immediate only if scored + ordered

- Crushing enteric/sustained = dose dump + ulcer/fail.
- IM in shock/hypotension → erratic; prefer IV/IO.
- Sublingual needs moist mouth + sit + no swallow/drink 2 min.

★ **Clinical Pearl:** Pearl: **Upright + water + stay 30 min** after oral Bisphosphonate/NSAID — oesophagus burn block.🔪 **NORCET Drill:** Mini-MCQ: Bypasses first-pass? (a) Oral (b) **Sublingual** (c) Enteric — Ans (b).**FATAL TRAP:** Crushing enteric/sustained tabs. Crush only plain immediate-release.**NORCET HIT:** Fast-IV, slow-oral, local-topical.

PRESCRIPTION PARTS + ABBREVIATIONS

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V2 DENSE

Core: **Patient-date-Rx-sign** complete; generic + strength + form + route + freq + duration + quantity; no banned shorts.

Patient + date	Name + age/weight + ID + date; allergy + diagnosis + pregnancy
Rx line	Generic + strength + form + route ; tall-man LASA
Directions	Dose-freq-duration-quantity + stop/review date
Instructions	Food + interactions + warning + follow-up + signature/reg
Abbreviations	BD/TDS/QID, AC/PC, HS, SOS, PRN ; no U/IU/QD/trailing zero
Verbal order	Repeat + write + read-back + countersign fast

- U = units confused as 0; QD = daily vs QID; trailing zero 5.0 = 50.
- LASA pair storage apart + alert + double.
- Verbal only emergency + two identifiers + document now.

★ **Clinical Pearl:** Pearl: **3 checks (pull-pour-bedside) + 5 rights + allergy band** before every administration.🔪 **NORCET Drill:** Mini-MCQ: Banned? (a) BD (b) **U + trailing zero** (c) HS — Ans (b).**FATAL TRAP:** U/IU/QD/trailing zero. Write units full, no banned shorts.**NORCET HIT:** Who-What-How long-Signed.

WEIGHTS, MEASURES & CONVERSIONS

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V2 DENSE

Core: **g-mg-mcg x1000**; L-mL x1000; tsp 5, tbsp 15, oz 30; kg-lb 2.2; % = g/100 mL; double-check decimal.

Mass	1 g = 1000 mg = 1,000,000 mcg ; 1 mg = 1000 mcg
Volume	1 L = 1000 mL; tsp 5, tbsp 15, cup ~240; drop ~0.05 mL
Weight	1 kg = 2.2 lb ; dosing kg; elderly/child reweigh
Percent	5% = 5 g/100 mL ; 1:1000 = 1 mg/mL
Worked	0.5 g = 500 mg ; 250 mcg = 0.25 mg; 70 kg = 154 lb
Safety	Label + decimal + second eyes high-alert; no mental-only

- mcg-mg 1000x slip kills (digoxin/insulin).
- % vs ratio confusion (adrenaline 1:1000 vs 1:10,000).
- Round + unit + pump library + handover.

★ **Clinical Pearl:** Pearl: **Leading zero (0.5) yes, trailing zero (5.0) never** — decimal safety rule.🔪 **NORCET Drill:** Mini-MCQ: 0.25 mg = ? mcg (a) 2.5 (b) **250** (c) 25 — Ans (b).**FATAL TRAP:** mg-mcg 1000x slip. Label + decimal + second eyes.**NORCET HIT:** Gram-mg-mcg x1000.

DOSE CALCULATION FORMULA

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V2 DENSE

Core: **D/H x Q** tablets/liquids; **mL/hr = vol/time**; gtt/min = vol x factor/time; weight mg/kg + max cap + double.

Tablet	Order 500 mg, have 250/tab → 500/250x1 = 2 tabs
Liquid	Order 375 mg, 250/5 mL → 375/250x5 = 7.5 mL
Infusion	500 mL/8h = 62.5 mL/hr ; pump + label + check
Drops	100 mL/hr x15/60 = 25 gtt/min
Weight	Paracetamol 15 mg/kg ; 20 kg = 300 mg + max/day cap
Double	Two nurses + read-back insulin/heparin/chemo/peds

- Unit mismatch (g vs mg) + decimal + pump misprogram.
- Max cap + allergy + organ + interaction before math.
- Document calc + verification + effect.

★ **Clinical Pearl:** Pearl: **Estimate first, calculate, sanity-check** (adult tab count 0.5-4 sane; 40 tabs = recheck).🔪 **NORCET Drill:** Mini-MCQ: 250 mg have, order 500 mg → ? (a) 0.5 (b) **2 tabs** (c) 4 — Ans (b).**FATAL TRAP:** Mental-only high-alert. Formula + label + verify.**NORCET HIT:** D-over-H-times-Q.

INFUSION + PEDIATRIC SAFETY

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V2 DENSE

Core: **Pump mL/hr + label + double**; weight today + BSA chemo; fluid overload watch (lungs/JVD/I-O).

Pump	Volume/time + drug + rate + double ; anti-free-flow + alarm
Weight	Today kg + mg/kg + max ; reweigh weekly/ill
BSA chemo	m2 + protocol + double + spill kit
Overload	Crackles/JVD/BP + slow/stop + sit + O2
Compat	Y-site chart + saline-only blood + no drug-mix
Handover	Rate + total + left + effect + plan + sign

- Gravity high-alert + free-flow flood.
- Adult dose scaled blind to child.
- Incompatible Y-mix precipitate embolus.

★ **Clinical Pearl:** Pearl: **1 mL/hr error x24h = 24 mL drift** — hourly volume check + pump lock + handover.🔪 **NORCET Drill:** Mini-MCQ: Safest high-alert infusion? (a) Gravity (b) **Pump + double + label** — Ans (b).**FATAL TRAP:** Adult dose for child. Weight + max + double.**NORCET HIT:** Weigh-Calc-Double.

PHARMACODYNAMICS: RECEPTORS

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V2 DENSE

Core: **Agonist on, antagonist block, partial mid**; synergism boost, tolerance creep; TI = toxic/effective.

Agonist	Salbutamol opens, morphine kills pain, insulin drops sugar
Antagonist	Naloxone reverses opioid, flumazenil benzo, VitK warfarin
Partial	Buprenorphine mid + ceiling + precipitated withdrawal
Synergism	Trimethoprim-sulfa, amox-clavulanate 1+1>2
Tolerance/dependence	Opioid/Benzo need more + never abrupt + taper
Therapeutic index	Narrow = digoxin/lithium/warfarin → levels + signs

- Antagonist as same effect.
- Abrupt stop dependence → seizure/storm/rebound.
- Narrow TI casual dosing.

★ **Clinical Pearl:** Pearl: **Antidote = antagonist logic** — block + support + monitor re-sedation/toxicity return.🔪 **NORCET Drill:** Mini-MCQ: Naloxone is? (a) Agonist (b) **Antagonist** — Ans (b).**FATAL TRAP:** Antagonist = same effect. Opposite — blocks.**NORCET HIT:** On-Block-Boost-Tolerate.

THERAPEUTIC INDEX + ADVERSE MAP

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V2 DENSE

Core: Side-predictable, allergy-immune, toxic-dose, idiosyncratic-odd ; allergy = stop + banner forever.

Side	Nausea/statin ache — warn + manage + continue if mild
Allergy	Rash-wheeze-drop → stop + adrenaline + banner + never rechallenge
Toxic	Digoxin vision/arrhythmia + low K; lithium thirst/tremor + level
Idiosyncratic	G6PD haemolysis, malignant hyperthermia — genetics + plan
Anaphylaxis	0.5 mg IM thigh + fluids + O2 + observe biphasic
Report	Effect + AE + vitals + lot + RCA + inform + document

- Rash as mere side → rechallenge death.
- Toxic levels casual + no hold.
- No allergy band + file update.

★ **Clinical Pearl:** Pearl: Allergy bracelet + file + handover + discharge card — four-place allergy lock.🔪 **NORCET Drill:** Mini-MCQ: Prior anaphylaxis drug again? (a) Cautious (b) **Never + banner** — Ans (b).**FATAL TRAP:** Rash as side effect. Allergy = stop + banner forever.**NORCET HIT:** Predictable-Immune-Toxic.

ADME I: ABSORPTION + BIOAVAILABILITY

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V2 DENSE

Core: **Oral-food-pH-motility, first-pass liver, shock-flow, sublingual bypass**; IM/SC need perfusion.

Food	Tetracycline-milk, thyroxine-food, metformin-food-ok + gaps
pH/motility	Antacid + enteric + diarrhoea/vomit cut absorb
First-pass	Morphine/propranolol low oral; SL/IV bypass
Perfusion	Shock/edema → poor IM/SC; use IV/IO
Topical/eye	Barrier + technique + nasolacrimal press + consent
Gap rule	Interacting pair 2-3h apart + fluids + sit

- IM vaccine in shock + no response.
- Antacid + antibiotic same minute.
- Crushed sustained + dump.

★ **Clinical Pearl:** Pearl: **Head injury + vomit + NPO → hold oral + IV + reassess swallow** before any PO drug.🔪 **NORCET Drill:** Mini-MCQ: Poor shock route? (a) IV (b) **IM/SC** — Ans (b).**FATAL TRAP:** IM in shock. IV/IO when perfusion poor.**NORCET HIT:** Route-Food-Liver-Flow.

ADME II: DISTRIBUTION + PROTEIN

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V2 DENSE

Core: **Albumin reservoir; free acts**; low albumin/pregnancy/old displace → toxicity at normal total; BBB/placenta selective.

Albumin low	Phenytoin/warfarin free up → toxicity ; check free + signs
Pregnancy	Free + fetal cross; benefit-risk + registry + folate
Elderly/renal	Fat + less water + slow clear → start low go slow
BBB	Meningitis + Parkinson cross needed; most excluded
Displacement	NSAID + warfarin + sulfa → bleed/kernicterus + monitor
Signs over numbers	Sedation/bleed/rash + vitals + levels decide

- Total normal = safe.
- Highly bound + displacer casual combo.
- Fall + bleed + confusion missed.

★ **Clinical Pearl:** Pearl: **Two highly-bound drugs + low albumin = free storm** — stagger + monitor + adjust.🔪 **NORCET Drill:** Mini-MCQ: Acts form? (a) Bound (b) **Free** — Ans (b).**FATAL TRAP:** Total level normal = safe. Free + signs decide.**NORCET HIT:** Bound waits, free acts.

METABOLISM + HALF-LIFE

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V2 DENSE

Core: **CYP inducers down, inhibitors up**; $t_{1/2} \rightarrow$ steady 4-5 halves; smoking/alcohol/grapefruit matter.

Inducer	Rifampicin/smoking/carbamazepine $\rightarrow$ OCP/failure + need more
Inhibitor	Ketoconazole/grapefruit/erythro $\rightarrow$ statin/toxic + avoid/reduce
Half-life	50% fall; steady 4-5 halves; loading vs maintenance
Liver fail	Jaundice/coagulopathy/encephalopathy $\rightarrow$ avoid + reduce + monitor
Counsel	Smoke/alcohol/herbal + adherence + interaction card
Missed dose	Half-life + window + never double high-alert + inform

- Rifampicin + OCP without barrier.
- Grapefruit + statin daily.
- Missed double-up narrow TI.

★ **Clinical Pearl:** Pearl: **Smoking stop = theophylline/clozapine levels jump** — pre-empt reduce + monitor.🔪 **NORCET Drill:** Mini-MCQ: Inducer does levels? (a) Up (b) **Down** — Ans (b).**FATAL TRAP:** Stop interacting pair blind. Ask + check + adjust.**NORCET HIT:** Induce-down, inhibit-up.

EXCRETION + RENAL/LIVER ADJUST

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V2 DENSE

Core: **GFR-dose, LFT-choice, levels-hydration**; nephro/ototoxic need trough + fluids + avoid NSAID/contrast stack.

GFR low	Extend interval/cut dose; aminoglycoside/vanco/digoxin watch
Liver fail	Avoid hepatotoxic + paracetamol cap + coagulopathy + encephalopathy
Nephrotoxic stack	NSAID + ACEI + diuretic + contrast = triple whammy + AKI
Hydration	2-3 L unless cardiac/renal cap + I-O + weights
Levels	Trough + peak + timing + signs + allergy + document
Elderly	Creatinine lies (low muscle) → eGFR + start low

- Full dose renal fail + no levels.
- Contrast + NSAID + dehydration stack.
- Creatinine normal = kidney fine in frail.

★ **Clinical Pearl:** Pearl: **Triple whammy (ACEI + diuretic + NSAID) + sick day = hold + fluids + GP call** — sick-day rules.🔪 **NORCET Drill:** Mini-MCQ: Triple whammy is? (a) Safe (b) **AKI stack — hold + hydrate** — Ans (b).**FATAL TRAP:** Full dose in renal failure. Adjust + monitor + hydrate.**NORCET HIT:** Filter-Function-Fix dose.

FACTORS: AGE/WEIGHT/PREGNANCY

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V2 DENSE

Core: **Old-young slow, weight-max, pregnancy-registry, G6PD-gene** ; start low go slow + fall + counsel.

Neonate	Immature liver/kidney + kernicterus sulfa + grey baby chloramphenicol — weight + dilute
Elderly	Beers + fall + confusion + renal + polypharmacy review + pill box
Weight/BSA	mg/kg + m2 chemo + ideal vs actual + cap + double
Pregnancy	Benefit-risk + folate + registry + ACEI/warfarin/valproate ban
Lactation	Hale + infant watch + pump-dump plan + timing feeds
Genetics	G6PD + HLA + porphyria + MH family + alert card

- Adult dose to neonate/old.
- Valproate/warfarin/ACEI in pregnancy.
- G6PD trigger + fava + infection.

★ **Clinical Pearl:** Pearl: **Pregnancy drug card: indication + gestation + dose + registry + folate + USG plan** filed + explained.🔪 **NORCET Drill:** Mini-MCQ: G6PD avoid? (a) Paracetamol (b) **Primaquine/dapsone** — Ans (b).**FATAL TRAP:** Same adult dose all. Individualize + monitor.**NORCET HIT:** Age-Weight-Gene-Gestation.

INTERACTIONS + FOOD-DRUG PAIRS

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V2 DENSE

Core: Warfarin-NSAID bleed, tetra-milk fail, MAOI-cheese crisis, metro-alcohol sick, K-supplement-ACEI arrest.

Warfarin + NSAID/garlic	Bleed + INR + ulcer → avoid + PPI + INR + soft
Tetra/quinolone + milk/antacid/iron	Chelate fail → gap 2-3h + sun + tendon
MAOI + cheese/wine	Tyramine crisis HTN + stroke → diet card + avoid
Metronidazole + alcohol	Disulfiram vomit/flush → zero + 72h + warn
ACEI + K/spironolactone	HyperK arrest → K + ECG + low-salt-substitute warn
Herbal + OTC	St John + OCP/transplant fail; ginkgo + bleed + ask all

- OTC/herbal as harmless.
- Cheese + MAOI casual party.
- Salt substitute (KCl) + ACEI + arrest.

★ **Clinical Pearl:** Pearl: **Brown bag review: bring all boxes + herbals + OTC every visit** — reconcile + de-prescribe + teach.🔪 **NORCET Drill:** Mini-MCQ: MAOI + cheese? (a) Fine (b) **Crisis — avoid** — Ans (b).**FATAL TRAP:** OTC/herbal as safe. Ask every drug + food.**NORCET HIT:** Pair-Pause-Counsel.

LAWS, IP & SCHEDULES

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A+**V2 DENSE****Core:** IP standards, H-Rx, H1-register, X-lock, NDPS-count ; rational right-review-stop + prescription audit.

IP	Identity-purity-strength book + batch + expiry + storage
H	Rx only + pharmacist file + counsel + no OTC
H1	Register + ID + 3rd-gen Ab/pregabalin + inspect
X/NDPS	Double-lock + count + register + report loss + witness waste
Rational	Indication + culture + allergy + cost + adherence + stop date
Audit	Legible + generic + banned-short + error + feedback

- H1/X as OTC + no register.
- No count/witness waste narcotic.
- Irrational FDC + antibiotic forever.

★ **Clinical Pearl:** Pearl: Narcotic count start-end + handover + incident + DPO call — loss = legal emergency.🔪 **NORCET Drill:** Mini-MCQ: H1 needs? (a) OTC (b) Rx + register — Ans (b).**FATAL TRAP:** X/H1 as OTC. Rx + register + count.**NORCET HIT:** Rx-Register-Restrict.

RATIONAL USE + P-DRUG HABIT

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V2 DENSE

Core: **P-drug per disease + allergy + organ + cost**; start-review-stop 48-72h; teach + diary + follow-up.

P-drug	Efficacy + safety + cost + availability + adherence winner
Start	Culture + diagnosis + allergy + renal + pregnancy + consent
Stay	Effect + AE + levels + adherence + diary + visit
Stop/switch	Narrow + IV-oral + date + handover + counsel
Teach	Dose-time-food + miss + store + red flags + card
System	Formulary + audit + feedback + de-prescribe + report

- Broad endless + no review.
- No teach + no diary + lost follow-up.
- Polypharmacy + Beers + fall.

★ **Clinical Pearl:** Pearl: **De-prescribe round: indication? benefit? harm? stop/taper + monitor + document** monthly elderly.🔪 **NORCET Drill:** Mini-MCQ: Rational review? (a) Never (b) **48-72h + stop date** — Ans (b).**FATAL TRAP:** Broad forever. Culture-first + stop date.**NORCET HIT:** Right-Review-Stop.

NURSING RESPONSIBILITY BUNDLE

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A+**V2 DENSE****Core: 3 checks + 5 rights + band + consent + teach + document + error-RCA** ; never hurry high-alert.

3 checks	Pull + pour + bedside + expiry + label + pump
5 rights + 3	Patient-drug-dose-time-route + documentation + reason + response
Band/consent	Two IDs + allergy + pregnancy + consent + site mark
Teach	Effect + AE + food + miss + store + red + card + diary
Document	Time-dose-route-site-effect-vitals + sign + handover
Error	Safe patient + vitals + inform + bag + report + RCA + learn

- Bedside check skip + similar names.
- No effect/AE charting + lost handover.
- Hide error + blame + no fix.

★ **Clinical Pearl:** Pearl: **High-alert vest + quiet zone + no-talk pouring + second eyes** cuts error most.🔪 **NORCET Drill:** Mini-MCQ: Checks when? (a) Once (b) **Pull-pour-bedside** — Ans (b).**FATAL TRAP:** Bedside check skip. Pull-pour-bedside + sign.**NORCET HIT:** Check-Scan-Sign.

BATCH 1 BLITZ V2

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V2 DENSE

Core: **Base locked denser** : PK/PD, routes, Rx, x1000, D/HxQ, agonist/antagonist, induce/inhibit, pairs, schedules, rational + rights.

PK vs PD	Body-drug vs drug-body
Bypass first-pass	Sublingual/IV
Never crush	Enteric/sustained
Banned	U + trailing zero
0.5 g	500 mg
Math	D/HxQ + double
Naloxone	Antagonist
Inducer/inhibitor	Down/up
Warfarin bleed	NSAID avoid + INR
Metro	No alcohol 72h
H1	Rx + register
Review	48-72h + date
Checks	Pull-pour-bedside
Free acts	Free + signs

- Use: cover right, recall aloud, tick weak → reread that page.
- Next Batch2 V2: antiseptics + GI + respiratory denser.

★ **Clinical Pearl:** Pearl: **V2 = highlight + pearl + MCQ fills bottom half — no empty page.**🔪 **NORCET Drill:** Mini-MCQ: V2 adds? (a) Space (b) **Density + highlighter** — Ans (b).**FATAL TRAP:** Base locked denser. Template proof done.**NORCET HIT:** Density + highlighter wins.

ANTISEPTICS VS DISINFECTANTS MATRIX

Professional Nursing Coach • Antiseptics + GI + Respiratory

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BATCH 2

Core: Chlorhexidine-alcohol (skin), povidone (wound), hypochlorite (spill/floor), glutaraldehyde (scope); concentration + contact + rinse.

Agent	Use / Guard
Chlorhexidine 2%+alcohol	Pre-op skin + line prep, dry fully
Povidone 10%	Wound/mucosa, rinse, thyroid caution
Hypochlorite 1%/0.5%	Blood spill / floor, fresh daily
Glutaraldehyde 2%	Scope HLD 20-45 min, rinse sterile

FATAL TRAP: Scope in hypochlorite. Corrodes + no HLD — glut/OPA only.**NORCET HIT:** Skin-CHG, wound-povidone, spill-hypo, scope-glut.

EMETICS & ANTIEMETICS

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BATCH 2

Core: Ondansetron (QT), metoclopramide (extrapyramidal), doxylamine (pregnancy), dexamethasone adjunct; cause first.

Drug	AE / Watch
Ondansetron	QT + headache + constipation; ECG
Metoclopramide	Dystonia/tardive; young caution
Doxylamine+B6	First-line pregnancy emesis

FATAL TRAP: Mask obstruction/ICP vomiting. Exclude surgical/neuro cause.**NORCET HIT:** Cause-Antiemetic-Hydrate.

LAXATIVES: BULK TO STIMULANT

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BATCH 2

Core: Bulk (psyllium-water), osmotic (lactulose/PEG), stimulant (bisacodyl short), softener (docusate); fluids + move + fiber first.

Class	Use / Ban
Bulk	Chronic + elderly, with water
Osmotic	Hepatic encephalopathy lactulose
Stimulant	Short prep, cramps/electrolyte

FATAL TRAP: Stimulant daily long. Habit + cathartic colon — bulk + cause.**NORCET HIT:** Fiber-Water-Move first.

ANTACIDS, H2, PPI, PROTECTANTS

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BATCH 2

Core: Antacid instant + interactions; H2 (ranitidine-type) night; PPI (omeprazole) 30 min breakfast; sucralfate empty; H.pylori triple.

Class	Timing / Pair
Antacid	1-3h post-meal + gap 2h others
H2 blocker	Night/ BD, renal adjust
PPI	30 min before breakfast, clopidogrel check
Sucralfate	Empty stomach, binds ulcer

FATAL TRAP: Tetracycline + antacid together. Gap 2-3h or fail.**NORCET HIT:** Instant-Timing-Gap.

H. PYLORI TRIPLE + ULCER CARE

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BATCH 2

Core: PPI + clarithromycin + amoxicillin/metronidazole 14d per current protocol; adherence + no NSAID/alcohol/smoke.

Need	Do
Confirm	Test + adherence + allergy
Protect	Sucralfate/misoprostol (pregnancy ban)
Warn	NSAID + alcohol + smoke stop

FATAL TRAP: NSAID continues silently. Stop/review + protect.**NORCET HIT:** Kill-protect-stop NSAID.

ORS + ANTIDIARRHOEALS

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BATCH 2

Core: Fluid-first: ORS after every stool + zinc kids; antimotility (loperamide) never in bloody/fever/kids dysentery.

Fluid	Rule
ORS	200-400 mL adult per stool + reassess
Zinc kids	10-20 mg 10-14d per protocol
Loperamide	No blood/fever/child invasive

FATAL TRAP: Drug before fluid. ORS + cause + red flags.**NORCET HIT:** Fluid-first, drug-cautious.

RELIEVER VS CONTROLLER ASTHMA

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BATCH 2

Core: Reliever SABA (salbutamol tremor/tachy) PRN; controller ICS/LABA daily + rinse; anticholinergic (ipratropium) + theophylline narrow.

Role	Drug / AE
Reliever	Salbutamol: tremor + tachy + K low
Controller	Budesonide: thrush/hoarse — rinse
Add-on	Theophylline: narrow, interaction, level

FATAL TRAP: Reliever daily overuse. Controller daily + action plan.

NORCET HIT: Blue-relief, brown-prevent.

INHALER + NEBULIZER TECHNIQUE

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BATCH 2

Core: MDI shake-prime-exhale-seal-inhale slow-hold 10s-rinse; spacer for kids/elderly; nebulizer mask + O2 + clean.

Step	Check
MDI	Shake + prime + slow + hold
Spacer	Seal + single puff + breathe
After ICS	Rinse-spit (thrush block)

FATAL TRAP: Fast puff + no hold. Slow-hold-rinse.**NORCET HIT:** Shake-Seal-Slow-Hold-Rinse.

DECONGESTANTS + ANTIHISTAMINES

Professional Nursing Coach • Antiseptics + GI + Respiratory

A+

BATCH 2

Core: Pseudoephedrine (HTN/BPH caution, short); 1st-gen (chlorpheniramine sleepy) vs 2nd-gen (cetirizine less); drying + urinary hold.

Class	Caution
Decongestant	HTN/glaucoma/BPH + rebound >3d
1st-gen AH	Sedation + fall + dry
2nd-gen AH	Less sleepy + renal adjust

FATAL TRAP: Drive on 1st-gen night dose. No drive + fall watch.**NORCET HIT:** Short-decongest, sleepy-first-gen.

EXPECTORANTS, ANTITUSSIVES, MUCOLYTICS

Professional Nursing Coach • Antiseptics + GI + Respiratory

A+

BATCH 2

Core: Guaifenesin + fluids; dextromethorphan dry cough only; acetylcysteine/bromhexine + hydration + physio.

Cough	Pick
Wet/productive	Expectorant + fluids + physio
Dry/annoying	Suppress short + cause check
Thick plug	Mucolytic + hydration

FATAL TRAP: Suppress productive cough. Clear + hydrate, suppress dry.**NORCET HIT:** Wet-clear, dry-calm.

GI NURSING TRAPS BANK

Professional Nursing Coach • Antiseptics + GI + Respiratory

A+

BATCH 2

Core: NSAID ulcer silent, elderly constipation impaction, bloody diarrhoea antibiotic mask, ORS mixing error.

Trap	Fix
NSAID + no protect	Risk score + PPI + review
Impaction	Rectal + disimpact + bulk
ORS salty	Exact litre + taste + discard 24h

FATAL TRAP: ORS eyeball mix. Exact water + 24h discard.**NORCET HIT:** Score-Protect-Mix exact.

RESPIRATORY TRAPS BANK

Professional Nursing Coach • Antiseptics + GI + Respiratory

A+

BATCH 2

Core: ICS without rinse thrush, theophylline toxicity (seizure/arrhythmia), SABA overuse hides flare, O₂ + CO₂ retainers.

Trap	Fix
Thrush	Rinse-spit + spacer + check
Theophylline	Level + no smoking shift + ECG
CO ₂ retainer	Targeted O ₂ + ABG + watch

FATAL TRAP: High O₂ all COPD. Target sats + ABG per protocol.**NORCET HIT:** Rinse-Level-Target.

CONSTIPATION PROTOCOL PAGE

Professional Nursing Coach • Antiseptics + GI + Respiratory

A+

BATCH 2

Core: Fiber + 2-3 L + walk + toilet time; bulk first, osmotic next, stimulant short; impaction manual + enema ordered.

Step	Do
Prevent	Fiber + fluids + move
Treat	Bulk → osmotic → short stimulant
Impaction	Assess + ordered disimpact

FATAL TRAP: Stimulant first chronic. Bulk + cause first.**NORCET HIT:** Prevent-Bulk-Short.

VOMITING + DEHYDRATION PROTOCOL

Professional Nursing Coach • Antiseptics + GI + Respiratory

A+

BATCH 2

Core: ABC + cause (obstruction/ICP/DKA/pregnancy) + vitals/I/O + ORS/IV + antiemetic + reassess + red-flag admit.

Red flag	Act
Blood/bile + pain	Surgical + NPO + NG
DKA/pregnancy	Glucose/ketone + OB + fluids
No urine/lethargy	Admit + IV + labs

FATAL TRAP: Antiemetic + discharge febrile bloody diarrhoea. Admit + culture.**NORCET HIT:** Cause-Fluid-Reassess.

COUGH TRIAGE PAGE

Professional Nursing Coach • Antiseptics + GI + Respiratory

A+

BATCH 2

Core: Red: stridor/drool, chest indrawing, SpO2 low, haemoptysis, TB contact; routine vs urgent split + inhaler check.

Sign	Go
Stridor/drool	Airway + upright + no poke
Wheeze + talk full	Reliever + spacer + review
TB cough 2w	Mask + sputum + notify

FATAL TRAP: Antitussive for stridor. Airway first.**NORCET HIT:** Triage-Technique-Follow.

NEBULIZATION NURSING STEPS

Professional Nursing Coach • Antiseptics + GI + Respiratory

A+

BATCH 2

Core: Dose + saline + O2 flow + mask seal + sit up + stay + vitals + clean + document + technique recheck.

Step	Do
Prepare	Drug + saline + expiry + allergy
Give	Sit + seal + breathe + watch
After	Rinse + clean + chart + plan

FATAL TRAP: Leave alone running. Stay + vitals + clean.**NORCET HIT:** Prepare-Stay-Clean.

BATCH 2 BLITZ I

Professional Nursing Coach • Antiseptics + GI + Respiratory

A+**BATCH 2****Core:** Antiseptic picks, emesis, laxative order, ulcer timing, ORS, reliever/controller, inhaler, decongestants.

Q	A
Skin prep	CHG-alcohol dry
Scope	Glut 20-45 min
Spill	1% hypo
Pregnancy emesis	Doxylamine+B6
Laxative first	Bulk + water
Ulcer timing	PPI 30 min breakfast
Gap	2h antacid
ORS rule	Fluid-first
Reliever	SABA PRN

FATAL TRAP: Half blitz done.**NORCET HIT:** GI + airway half.

BATCH 2 BLITZ II

Professional Nursing Coach • Antiseptics + GI + Respiratory

A+

BATCH 2

Core: Controller rinse, theophylline, sedation, wet/dry cough, traps, protocols, next CVS.

Q	A
Controller guard	Rinse-spit
Narrow	Theophylline level
Sleepy	1st-gen AH
Wet cough	Expectorant + fluids
Dry	Suppress short
COPD O2	Target + ABG
Impaction	Ordered disimpact
Red vomit	Surgical + NPO
Next	CVS/blood

FATAL TRAP: Batch 2 complete.**NORCET HIT:** Continue CVS.

HAEMATINICS: IRON, FOLATE, B12

Professional Nursing Coach • CVS & Blood I

A+

BATCH 3

Core: Iron empty + VitC, no tea/antacid; folate pregnancy; B12 IM if pernicious; Hb + ferritin + retic monitor.

Drug	Give / Watch
Ferrous oral	Empty + VitC, black stool normal, constipation
Iron IV	Test + anaphylaxis kit + slow
Folate	Pregnancy NTD block, with B12 check
B12 IM	Monthly if intrinsic lack, neuro review

FATAL TRAP: Milk/tea with iron. Gap + VitC + adherence.**NORCET HIT:** Empty-VitC-Black normal.

AUTONOMIC: CHOLINERGIC VS ANTICHOLINERGIC

Professional Nursing Coach • CVS & Blood I

A+

BATCH 3

Core: Cholinergic SLUDGE (neostigmine myasthenia); anticholinergic dry-hot-blind-mad (atropine pre-op bradycardia).

System	Signs / Drug
Cholinergic on	Salivate-lacrimate-urinate-defecate + miosis
Anticholinergic on	Dry-hot-flush-mydriasis-retention
Myasthenia	Neostigmine improves, Tensilon test

FATAL TRAP: Atropine for SLUDGE. Opposite — atropine dries.**NORCET HIT:** Wet-cholinergic, dry-atropine.

ADRENERGIC VS ANTIADRENERGIC

Professional Nursing Coach • CVS & Blood I

A+

BATCH 3

Core: Adrenaline (anaphylaxis IM thigh) vs beta-blocker (slow + mask hypo); alpha pressors vs alpha-blocker hypotension.

Drug	Use / AE
Adrenaline	Anaphylaxis 0.5 IM thigh + fluids
Noradrenaline	Shock infusion + extravasation watch
Beta-blocker	HTN/angina + bradycardia + bronchospasm

FATAL TRAP: Beta-blocker abrupt stop. Taper + HR/BP + sugar.**NORCET HIT:** Press-Adrenaline, slow-Beta.

HEART FAILURE + VASODILATORS

Professional Nursing Coach • CVS & Blood I

A+

BATCH 3

Core: ACEI cough/K up, ARB no cough, diuretic K/fluids, digoxin narrow (vision/arrhythmia), nitro preload caution.

Drug	Monitor / Hold
Enalapril/losartan	K + creatinine + cough/dry
Furosemide	I/O + K + BP + ototoxic fast IV
Digoxin	HR + K + vision + level 0.8-2

FATAL TRAP: Digoxin with low K + vomit. Hold + K + ECG + level.**NORCET HIT:** K-BP-HR-Level every HF drug.

DIGOXIN TOXICITY CARD

Professional Nursing Coach • CVS & Blood I

A+

BATCH 3

Core: Anorexia-nausea-vision yellow-arrhythmia + low K/Mg + renal; hold + K/Mg + ECG + Fab per protocol + never push K blind hyperK.

Trigger	Fix
Low K/diuretic	Replace + monitor + hold
Renal/old	Reduce + level + ECG
Toxic signs	Hold + notify + Fab if severe

FATAL TRAP: Calcium IV in toxic heart. Fatal stone — avoid + specialist.

NORCET HIT: Yellow-vision-hold-level.

ANTIANGINALS: NITRATES/BB/CCB

Professional Nursing Coach • CVS & Blood I

A+

BATCH 3

Core: Nitrate headache/hypotension + tolerance + no sildenafil; BB slow; CCB ankle/flush/constipation; chest pain protocol.

Class	Key / Ban
Nitroglycerin SL	Sit + 5-min x3 + call + headache
BB/CCB	HR/BP + no abrupt + grapefruit CCB
Sildenafil + nitrate	Fatal drop — 24-48h gap

FATAL TRAP: Nitrate + sildenafil together. Ask + gap + warn.**NORCET HIT:** Sit-SL-Call, no blue+nitro.

NITRO ADMINISTRATION STEPS

Professional Nursing Coach • CVS & Blood I

A+

BATCH 3

Core: Sit + BP/HR + pain score + SL + 5-min reassess x3 + ECG + aspirin/heparin per ACS + admit + handover.

Step	Do
Before	Sit + vitals + ECG + allergy
Give	SL + 5-min + pain + vitals
After	Respond/call + document + monitor

FATAL TRAP: Stand + drive after nitro. Sit + observe + escort.**NORCET HIT:** Sit-SL-Stay.

ANTIARRHYTHMICS OVERVIEW

Professional Nursing Coach • CVS & Blood I

A+

BATCH 3

Core: Adenosine rapid SVT (warn asystole seconds) + amiodarone slow (thyroid/lung/skin) + BB/CCB rate + QT watch + Mg/K fix.

Drug	Watch
Adenosine IV fast	Brief asystole + flush + warn
Amiodarone	Thyroid + lung + blue skin + QT
Diltiazem/verapamil	HR/BP + constipation + HF

FATAL TRAP: Verapamil + BB IV together. Separate + ICU + calcium ready.**NORCET HIT:** Fast-adeno, slow-amio.

QT + BRADYCARDIA TRAPS

Professional Nursing Coach • CVS & Blood I

A+

BATCH 3

Core: QT drugs (macrolide/antipsychotic/amio) + low K/Mg = torsades; bradycardia + block + fall + pacemaker plan.

Risk	Fix
Long QT	ECG + K/Mg + stop culprit
HR<50 symptomatic	Atropine + pacing + cause
Fall on BB	HR/BP + med review

FATAL TRAP: QT blind polypharmacy. ECG + electrolytes + list.**NORCET HIT:** ECG-K-Mg-List.

HTN CLASSES I: ACEI/ARB/CCB

Professional Nursing Coach • CVS & Blood I

A+

BATCH 3

Core: ACEI dry cough + K/creatinine; ARB no cough; CCB ankle/flush/gum; pregnancy ban ACEI/ARB.

Class	AE / Ban
ACEI	Cough + K + creatinine + pregnancy X
ARB	No cough + same K/pregnancy
CCB	Ankle + flush + constipation

FATAL TRAP: ACEI in pregnancy. Stop + switch + counsel.**NORCET HIT:** Cough-K-Creatinine-Ankle.

HTN CLASSES II: DIURETICS/BB/CENTRAL

Professional Nursing Coach • CVS & Blood I

A+

BATCH 3

Core: Thiazide K/Na/glucose; loop powerful + ototoxic; BB slow/mask; methyldopa pregnancy HTN; clonidine rebound.

Class	Watch
Thiazide	K + Na + gout + glucose
Loop	I/O + K + hearing fast IV
Clonidine stop	Rebound HTN — taper

FATAL TRAP: Clonidine abrupt stop. Taper + BP + rescue.**NORCET HIT:** Salt-Sugar-Hearing-Rebound.

HTN EMERGENCY VS URGENCY

Professional Nursing Coach • CVS & Blood I

A+

BATCH 3

Core: Emergency (organ damage) IV titrate + ICU + neuro watch; urgency oral + 24-48h lower + cause + adherence.

Type	Act
Emergency + chest/stroke/eclampsia	ICU + IV + organ + no rapid drop
Urgency no damage	Oral + rest + cause + follow-up
Adherence	Pill box + side + cost + visit

FATAL TRAP: Rapid normalize emergency. Controlled 10-20% MAP first hour max per protocol.**NORCET HIT:** Organ decides speed.

SHOCK FLUIDS + PRESSORS LINK

Professional Nursing Coach • CVS & Blood I

A+

BATCH 3

Core: Crystalloid first + blood/causes; noradrenaline/dopamine/dobutamine titrate + extravasation + perfusion (urine/mental/CRT).

Shock	Support
Hypovolemic	Blood/fluids + cause + warm
Cardiogenic	Inotrope + O2 + ICU + echo
Septic	Fluids + cultures + Ab + lactate

FATAL TRAP: Pressor without fluids. Fill + cause + press.**NORCET HIT:** Fill-Cause-Press-Perfuse.

BLEED-BP-HR BUNDLE

Professional Nursing Coach • CVS & Blood I

A+**BATCH 3****Core:** Postural vitals + ECG + K/Mg + renal + allergy + adherence + home BP diary + fall plan.

Check	Do
Vitals	Sit-stand + HR + ECG
Labs	K + creatinine + lipids + sugar
Teach	Salt + smoke + pills + diary

FATAL TRAP: No home diary. Diary + review + adjust.**NORCET HIT:** Vitals-Labs-Diary.

BATCH 3 BLITZ I

Professional Nursing Coach • CVS & Blood I

A+**BATCH 3****Core:** Iron empty, SLUDGE/dry, adrenaline thigh, HF K/BP, digoxin vision, nitro sit, adeno fast, amio slow.

Q	A
Iron +	Empty + VitC
SLUDGE drug	Neostigmine
Dry-hot	Atropine
Anaphylaxis	Adrenaline 0.5 thigh
ACEI cough	Switch ARB
Digoxin see	Yellow vision
Nitro + blue	Fatal gap
SVT fast	Adenosine warn
Slow thyroid	Amiodarone

FATAL TRAP: Half done.**NORCET HIT:** CVS half.

BATCH 3 BLITZ II

Professional Nursing Coach • CVS & Blood I

A+

BATCH 3

Core: HTN bans, diuretic watch, emergency speed, shock fill, diary, next anticoag.

Q	A
Pregnancy ban	ACEI/ARB
CCB ankle	Amlodipine
Thiazide loses	K + Na
Loop ears	Ototoxic fast
Central rebound	Clonidine taper
Emergency	Organ + ICU IV
Shock first	Fluids + cause
Home	BP diary
Next	Clot + lipid

FATAL TRAP: Batch 3 complete.**NORCET HIT:** Next anticoag.

HEPARIN VS WARFARIN VS DOAC

Professional Nursing Coach • Clot + Lipid + Endocrine + Skin

A+

BATCH 4

Core: Heparin fast IV/SC + aPTT + HIT + protamine; warfarin slow oral + INR 2-3 + VitK; DOAC fixed + renal + specific reversal per current protocol.

Drug	Monitor / Reverse
Heparin IV/SC	aPTT 1.5-2.5x + platelets (HIT) → protamine
Warfarin oral	INR 2-3 (value higher) → VitK/PCC
DOAC (e.g. dabigatran/riva)	Renal + bleed plan → idarucizumab/and exanet per protocol

FATAL TRAP: Bridging/INR blind surgery. Hold + plan + reverse ready.

NORCET HIT: Fast-heparin, slow-warfarin, fixed-DOAC.

ANTIPLATELETS + THROMBOLYTICS

Professional Nursing Coach • Clot + Lipid + Endocrine + Skin

A+

BATCH 4

Core: Aspirin + clopidogrel dual post-stent; thrombolytic (alteplase) stroke/MI window + bleed screen + neuro watch.

Drug	Guard
Aspirin/clopidogrel	Bleed + ulcer + surgery hold
Alteplase/tenecteplase	Window + BP + neuro + no poke
After lysis	Neuro + BP + bleed + no IM/NG trauma

FATAL TRAP: Lysis without bleed screen. Checklist + window + consent.**NORCET HIT:** Dual-platelet, window-lytic.

BLEEDING PRECAUTIONS BUNDLE

Professional Nursing Coach • Clot + Lipid + Endocrine + Skin

A+

BATCH 4

Core: Soft brush, electric razor, no IM/NSAID/alcohol, pressure + ice, fall-proof, INR diary, antidote chart bedside.

Risk	Fix
Cut/bleed	Pressure 10-15 min + ice + elevate
Fall/head	Neuro + CT + hold + reverse
Surgery/delivery	Hold plan + crossmatch + reverse

FATAL TRAP: IM + NSAID on warfarin. Oral + soft + hold plan.**NORCET HIT:** Soft-Pressure-Hold-Reverse.

COAGULANTS: VITK + TRANEXAMIC

Professional Nursing Coach • Clot + Lipid + Endocrine + Skin

A+

BATCH 4

Core: VitK slow IV (anaphylaxis fast) for warfarin bleed; tranexamic for PPH/menorrhagia/trauma per protocol + clot watch.

Drug	Give
VitK IV	Slow + allergy kit + INR
Tranexamic	Early + clot + renal
Blood + PCC	Type + vitals + reaction watch

FATAL TRAP: Fast IV VitK push. Slow + monitor + allergy ready.**NORCET HIT:** Slow VitK, early TXA.

HYPOLIPIDEMICS: STATINS CORE

Professional Nursing Coach • Clot + Lipid + Endocrine + Skin

A+

BATCH 4

Core: Atorvastatin night + LFT + muscle pain/CK + grapefruit + pregnancy ban + diabetes watch + adherence.

Check	Do
Baseline	LFT + lipids + sugar + pregnancy
Warn	Muscle pain + dark urine → hold + CK
Pair	Grapefruit + macrolide raise

FATAL TRAP: Statin in pregnancy. Stop + switch + counsel.**NORCET HIT:** Night-LFT-Muscle-No grapefruit.

SHOCK FLUIDS: CRYSTALLOID/COLLOID

Professional Nursing Coach • Clot + Lipid + Endocrine + Skin

A+

BATCH 4

Core: NS/RL first + blood/causes; albumin/colloid selective; pressor titrate + urine/mental/CRT + lactate.

Fluid	Use / Caution
NS/RL	First + reassess + overload watch
Blood	Loss + trigger + reaction
Albumin	Selective + cost + allergy

FATAL TRAP: Fluids without reassess. Bolus-reassess-lungs.**NORCET HIT:** Fill-Reassess-Perfuse.

INSULIN TYPES TIMELINE

Professional Nursing Coach • Clot + Lipid + Endocrine + Skin

A+

BATCH 4

Core: Rapid (lispro minutes) / short (regular 30-60) / intermediate (NPH cloudy) / long (glargine clear, no mix); peak = hypo time.

Type	Peak / Rule
Lispro/aspart	15-30 min + meal now
Regular	2-4h + 30 min before
NPH	4-8h cloudy, roll-mix
Glargine	Flat 24h, never mix

FATAL TRAP: Mix glargine with others. Never mix long-flat.**NORCET HIT:** Peak-hypo, cloudy-roll, flat-no mix.

INSULIN ADMIN + STORAGE

Professional Nursing Coach • Clot + Lipid + Endocrine + Skin

A+

BATCH 4

Core: Abdomen fastest + rotate + no massage; 2-8C stock, room in-use 28d; double + meal confirmed + hypo kit.

Step	Do
Site	Abdomen + rotate + 1-inch gap
Dose	Double + meal + sugar + hold if NPO
Store	Fridge stock, room 28d, no freeze

FATAL TRAP: Sliding only without basal. Basal-bolus + sick rules.**NORCET HIT:** Rotate-Double-Meal-28d.

HYPOGLYCEMIA RESCUE

Professional Nursing Coach • Clot + Lipid + Endocrine + Skin

A+

BATCH 4

Core: Shaky-sweaty-hungry → 15 g fast carb + recheck 15 min x3; unconscious → glucagon/IV dextrose + airway + cause.

Level	Act
Awake	15 g juice/glucose + recheck
Drowsy/NPO	Glucagon/IV D + airway + monitor
After	Cause + reduce + teach + ID

FATAL TRAP: Chocolate + drive. Fast carb + recheck + cause.**NORCET HIT:** 15-15-15 + cause.

ORAL AGENTS: METFORMIN TO SGLT2

Professional Nursing Coach • Clot + Lipid + Endocrine + Skin

A+

BATCH 4

Core: Metformin first + B12/lactate + hold contrast/surgery; SUs hypo/weight; SGLT2 UTI/DKA + foot; gliptins pancreatitis.

Class	Watch
Metformin	Lactate + B12 + hold dye/surgery
SU (glimepiride)	Hypo + weight + elderly caution
SGLT2	UTI + DKA + volume + foot

FATAL TRAP: Metformin in shock/hypoxia. Hold + lactate + hydrate.**NORCET HIT:** First-metformin, hypo-SU, wet-SGLT2.

THYROID + ANTITHYROID

Professional Nursing Coach • Clot + Lipid + Endocrine + Skin

A+

BATCH 4

Core: Levothyroxine empty AM + iron/calcium gap + TSH; carbimazole rash/agranulocytosis (sore throat-fever-stop-CBC).

Drug	Rule
Levothyroxine	Empty + 4h gap + TSH
Carbimazole	Fever-sore throat → stop + CBC
Thyroid storm	Cool + fluids + BB + iodine per protocol

FATAL TRAP: Night levothyroxine with milk. Empty AM + gap.**NORCET HIT:** Empty-AM, fever-stop.

STEROIDS: BENEFIT VS TAPER

Professional Nursing Coach • Clot + Lipid + Endocrine + Skin

A+

BATCH 4

Core: Short burst vs long moon-face-sugar-bone-infection; never abrupt after 2 wks — taper + stress cover + vaccine live ban.

Use	Guard
Burst	Sugar + sleep + ulcer + mood
Long	Bone + eye + TB + growth + adrenal
Stop	Taper + stress dose + card

FATAL TRAP: Abrupt stop after weeks. Taper + card + cover.**NORCET HIT:** Taper-Card-Cover.

CALCIUM/VITD/CALCITONIN

Professional Nursing Coach • Clot + Lipid + Endocrine + Skin

A+

BATCH 4

Core: Calcium + VitD + sun + fall-proof; hypo-tingling-Chvostek vs hyper-stone-bone-pain; CKD/bisphosphonate upright.

State	Fix
Low (tingle/spasm)	Calcium + VitD + Mg + ECG
High (stones/pain)	Fluids + mobilize + cause
Bone drug	Upright + water + empty

FATAL TRAP: IV calcium fast push. Slow + ECG + Mg.**NORCET HIT:** Tingle-low, stone-high.

SKIN + TOPICAL SAFETY

Professional Nursing Coach • Clot + Lipid + Endocrine + Skin

A+

BATCH 4

Core: Thin skin + occlusion ↑ absorb; steroids fingertip + short + taper; scabicide neck-down + contacts + linen hot.

Topical	Rule
Steroid	Fingertip + short + face caution
Antifungal	Dry + 2 wks beyond clear
Scabies	Neck-down + contacts + linen

FATAL TRAP: Potent steroid face long. Mild-short-taper + moisturize.**NORCET HIT:** Thin-short-taper.

BATCH 4 BLITZ

Professional Nursing Coach • Clot + Lipid + Endocrine + Skin

A+**BATCH 4****Core:** Heparin-aPTT-protamine, warfarin-INR-VitK, dual platelet, lysis window, statin muscle, insulin peak, hypo 15, thyroxine AM, steroid taper.

Q	A
Heparin check	aPTT + HIT
Warfarin check	INR 2-3
DOAC guard	Renal + bleed plan
Dual after	Stent 12 mo
Lysis needs	Window + screen
Statin warn	Muscle + LFT
Rapid meal	Now
Glargine	Never mix
Hypo awake	15 g + recheck
Thyroxine	Empty AM
Steroid stop	Taper
Calcium low	Tingle
Scabies	Contacts + linen

FATAL TRAP: Batch 4 complete.**NORCET HIT:** Next antimicrobials.

ANTIBIOTIC FIRST: CULTURE + ALLERGY + STEWARDSHIP

Professional Nursing Coach • Antimicrobials + Antidotes

A+

BATCH 5

Core: Culture/sensitivity + allergy + indication before first dose; empiric → definitive 48-72h; spectrum + adherence + renal.

Before	Do
Culture	2 sets/samples before Ab
Allergy	Drug + food + band + ask
Review	48-72h de-escalate + stop date

FATAL TRAP: Broad forever without review. Culture-first + stop date.**NORCET HIT:** Culture-Allergy-Review.

PENICILLINS + CEPHALOSPORINS

Professional Nursing Coach • Antimicrobials + Antidotes

A+

BATCH 5

Core: Amoxicillin rash/diarrhoea; allergy + cross cephalosporin caution; anaphylaxis kit; renal adjust; no self-stop.

Class	Watch
Penicillin	Allergy + anaphylaxis + rash
Cephalo	Cross 10% + bleed (MTT) + alcohol (some)
Give	Test dose per protocol + observe

FATAL TRAP: Rechallenge after allergy. Banner + never again + alternative.**NORCET HIT:** Allergy-Banner-Observe.

AMINOGLYCOSIDES + MACROLIDES

Professional Nursing Coach • Antimicrobials + Antidotes

A+

BATCH 5

Core: Gentamicin oto/nephro + levels + hydration + once-daily; azithro QT + liver; erythro pregnancy-safe + GI upset.

Drug	Monitor
Gentamicin	Peak-trough + hearing + creatinine + fluids
Azithromycin	QT + liver + allergy
Erythro	GI + safe pregnancy + QT

FATAL TRAP: Loop + amino together deaf. Separate + levels + hydrate.**NORCET HIT:** Ears-Kidney-QT trio.

QUINOLONES + SULFA + TETRACYCLINE

Professional Nursing Coach • Antimicrobials + Antidotes

A+

BATCH 5

Core: Cipro tendon/CNS + theophylline up + sun; sulfa kernicterus + SJS + fluids; tetracycline teeth + milk gap + pregnancy ban.

Class	Ban / Gap
Quinolone	Tendon + kids/pregnancy caution + sun
Sulfa	Newborn + SJS + fluids + allergy
Tetracycline	<8 yrs + pregnancy + milk gap 2-3h

FATAL TRAP: Milk with tetracycline. Gap 2-3h + sun + tendon watch.**NORCET HIT:** Tendon-Teeth-Sun-Gap.

METRONIDAZOLE + ANAEROBES

Professional Nursing Coach • Antimicrobials + Antidotes

A+

BATCH 5

Core: Bitter + neuropathy + seizures high; disulfiram with alcohol — abstain during + 48-72h after; dark urine normal.

Counsel	Do
Alcohol	Zero during + 3 days after
Neuro	Tingling/seizure → hold + notify
Course	Complete + adherence

FATAL TRAP: Beer on metronidazole. Zero alcohol + warn + document.**NORCET HIT:** No alcohol 3 days.

ATT: HRZE TOXICITIES

Professional Nursing Coach • Antimicrobials + Antidotes

A+

BATCH 5

Core: H liver + neuropathy (B6); R orange + liver + OCP fail + uveitis; Z gout/liver; E vision (red-green) + monthly eye; DOT + adherence.

Drug	Key AE / Monitor
Isoniazid	Liver + neuropathy + B6
Rifampicin	Orange + liver + OCP + lenses
Pyrazinamide	Gout + liver + uric acid
Ethambutol	Optic + red-green + eye chart

FATAL TRAP: Stop ATT on orange urine. Normal — counsel + continue + monitor LFT/vision.**NORCET HIT:** Orange-normal, vision-stop, liver-watch.

MDT LEPROSY + ADHERENCE

Professional Nursing Coach • Antimicrobials + Antidotes

A+

BATCH 5

Core: Rifampicin + dapsone (haemolysis/G6PD) + clofazimine (brown skin); blister packs + 6/12 mo + no isolation on treatment.

Drug	Watch
Dapsone	Haemolysis + rash + G6PD
Clofazimine	Brown + dry + constipation
Adherence	Blister + DOT + stigma counsel

FATAL TRAP: Isolate treated leprosy. Non-infectious on MDT + counsel.**NORCET HIT:** Blister-Complete-No stigma.

MALARIA: ACT + PRIMAQUINE

Professional Nursing Coach • Antimicrobials + Antidotes

A+

BATCH 5

Core: Species + resistance per current NCVBDC; artemether-lumefantrine + primaquine radical (G6PD first); CQ retinopathy long.

Drug	Guard
ACT	Fatty food + QT + adherence 3d
Primaquine	G6PD + haemolysis + pregnancy caution
CQ/HCQ	Eye + QT + sugar

FATAL TRAP: Primaquine without G6PD. Test + counsel + follow Hb.**NORCET HIT:** Species-G6PD-Adherence.

ANTIFUNGALS + ART LINKS

Professional Nursing Coach • Antimicrobials + Antidotes

A+

BATCH 5

Core: Fluconazole liver/QT + amphi kidney/mg/K + infusion reaction; ART adherence >95% + OI prophylaxis + no sharing.

Drug	Monitor
Fluconazole	Liver + QT + pregnancy
Amphotericin	Kidney + K/Mg + chills + pre-hydrate
ART	Adherence + OI + stigma-free

FATAL TRAP: Stop ART when better. Lifelong + adherence +_supply.**NORCET HIT:** Adherence for life.

ANTIDOTES I: BLEED + BRADYCARDIA

Professional Nursing Coach • Antimicrobials + Antidotes

A+

BATCH 5

Core: Warfarin-VitK/PCC, heparin-protamine, DAB-and exanet per protocol; atropine-bradycardia, Ca-channel/Verapamil-calcium (caution digoxin toxic).

Toxin	Reverse
Warfarin bleed	VitK slow + PCC + hold
Heparin/HIT	Protamine + platelets + stop
Bradycardia	Atropine + pacing + cause

FATAL TRAP: Fast VitK push. Slow + allergy + INR.

NORCET HIT: Hold-Reverse-Monitor.

ANTIDOTES II: OPIOID + PARACETAMOL + DIGOXIN

Professional Nursing Coach • Antimicrobials + Antidotes

A+

BATCH 5

Core: Naloxone titrated (re-sedation watch); NAC within 8h paracetamol + LFT; digoxin Fab severe + K/Mg + no IV Ca toxic.

Overdose	Give + watch
Opioid slow	Naloxone titrate + re-sedation 2h
Paracetamol	NAC <8h + LFT + Rumack
Digoxin severe	Fab + hold + K/Mg + ECG

FATAL TRAP: Single naloxone + discharge. Observe re-sedation + cause.**NORCET HIT:** Titrate-Observe-Cause.

ANTIDOTES III: HYPO + ALLERGY + CHOLINERGIC

Professional Nursing Coach • Antimicrobials + Antidotes

A+

BATCH 5

Core: Glucose/glucagon hypo; adrenaline anaphylaxis thigh; atropine+pralidoxime organophosphate (SLUDGE + bronchorrhea) + decontaminate.

Crisis	First
Hypo unconscious	Glucagon/IV D + airway
Anaphylaxis	Adrenaline 0.5 thigh + fluids
OP poison	Atropine + pralidoxime + wash + ICU

FATAL TRAP: Antihistamine alone anaphylaxis. Adrenaline first.

NORCET HIT: Airway-Adrenaline-Atropine.

STEWARDSHIP + ALLERGY BUNDLE

Professional Nursing Coach • Antimicrobials + Antidotes

A+

BATCH 5

Core: Indication + culture + allergy + renal + pregnancy + stop date + handover + patient teach (complete, no share/OTC).

Bundle	Do
Start	Culture + allergy + dose
Stay	Levels + effect + AE + adherence
Stop/switch	Review 48-72h + narrow + date

FATAL TRAP: No stop date. Date + review + narrow.**NORCET HIT:** Start-Review-Stop.

REGIMEN BOXES (UPDATEABLE)

Professional Nursing Coach • Antimicrobials + Antidotes

A+**BATCH 5****Core:** ATT/MDT/malaria/ART change — write current NTEP/NLEP/NCVBDC/NACO regimen + date + source; exam qualifies with date.

Box	Fill current
ATT adult	HRZE per weight + DOT + date
MDT	PB 6 / MB 12 blister + date
Malaria/ART	Species/drug + adherence + date

FATAL TRAP: Frozen old regimen as current. Date-qualified + source.**NORCET HIT:** Check-date-cite.

BATCH 5 BLITZ

Professional Nursing Coach • Antimicrobials + Antidotes

A+

BATCH 5

Core: Culture-first, allergy banner, amino ears/kidney, sulfa newborn, metro alcohol, ATT orange/vision, MDT blister, ACT G6PD, antidotes trio.

Q	A
First dose need	Culture + allergy
Amino check	Levels + hearing + creatinine
Newborn ban	Sulfa (kernicterus)
Teeth ban	Tetracycline <8/preg
Metro +	No alcohol 3d
Orange	Rifampicin normal
Vision drug	Ethambutol stop
Leprosy packs	Blister 6/12
Malaria radical	Primaquine G6PD
Warfarin reverse	VitK/PCC
Opioid reverse	Naloxone titrate
PCM antidote	NAC <8h
Anaphylaxis	Adrenaline thigh

FATAL TRAP: Pharma I complete.**NORCET HIT:** Compendium next.

DNA + karyotype + audit! 46/23 pairs + gene locus + audit — code + count + audit!

DNA/gene +
code + verify

46/23 pairs +
karyotype + audit

Code + count
+ pride!

★ CODE — COUNTS AS

- DNA gene + chromosome + locus + verify. Ex: 46 = 23 pairs.
- Karyotype: count + sex + audit. Ex: XY male + XX female.

⚠ FATAL TRAP

Cell-only + 'gene asked' + code-miss + count-miss =
wrong + fail! Code + count + audit.

★ NORCET HIT

HIT: Division + Mendel next — halve + laws + audit!

► Code + count + locus + karyotype + sort?

Gene + 46 + audit.

Halve + laws + audit! Meiosis + 3 laws + audit — halve + laws + audit!

Meiosis halve +
gametes + verify

Dominance/split/
sort + laws

Halve + laws
+ pride!

▲ LAWS — HALVE AS

- Meiosis halves + gametes + variation + verify. Ex: sperm/ovum 23.
- Mendel: dominance + segregation + independent + audit. Ex: pea plants.

⚠ FATAL TRAP

Divide-only + 'Mendel asked' + halve-miss + law-miss =
wrong + fail! Halve + laws + audit.

★ NORCET HIT

HIT: Punnett + ratios next — square + 3:1 + audit!

► Halve + laws + pea + gamete + link?

Meiosis + Mendel + audit.

Square + 3:1 + audit! 9:3:3:1 + predict + audit — square + ratio + audit!

Square + fill +
predict + verify

3:1 + 9:3:3:1
+ ratios + audit

Square + ratio
+ pride!

◆ SQUARE — RATIOS AS

- Punnett: fill + predict + verify. Ex: Aa x Aa = 3:1.
- Dihybrid 9:3:3:1 + test cross + audit. Ex: carrier hunt.

⚠ FATAL TRAP

Box-only + 'cross asked' + fill-miss + ratio-miss = wrong
+ fail! Fill + ratio + audit.

★ NORCET HIT

HIT: Dominant disorders next — Huntington + 50 pct +
audit!

► Fill + ratio + predict + cross + math?

Punnett + ratios + audit.

One dose + 50 pct + audit! Huntington/Marfan + vertical + audit — one + half + audit!

One dose +
shows + verify

Huntington/Marfan +
50 pct + audit

One + half
+ pride!

Σ ONE — HALVES AS

- AD: one dose shows + vertical + 50 pct + verify.
- List: Huntington + achondroplasia + Marfan + neurofibromatosis + audit.

⚠ FATAL TRAP

Gene-only + 'dominant asked' + dose-miss + half-miss
= wrong + fail! Dose + half + audit.

★ NORCET HIT

HIT: Recessive next — two + quarter + audit!

► Dose + half + vertical + list + sort?

AD + 50 + audit.

Two + quarter + audit! CF/sickle/thalassemia + horizontal + audit — two + quarter + audit!

Two doses +
carrier + verify

CF/sickle/thal +
25 pct + audit

Two + quarter
+ pride!

★ TWO — QUARTERS AS

- AR: two doses + carriers hide + 25 pct + verify. Ex: cousins risk.
- List: CF + sickle + thalassemia + Tay-Sachs + PKU + audit.

⚠ FATAL TRAP

Gene-only + 'recessive asked' + dose-miss +
quarter-miss = wrong + fail! Dose + quarter + audit.

★ NORCET HIT

HIT: X-linked next — males + criss-cross + audit!

► Dose + quarter + carrier + list + sort?

AR + 25 + audit.

Males + criss-cross + audit! Hemophilia/colour + audit — males + cross + audit!

Males show +
carrier-mom + verify

Hemophilia/colour/
Duchenne + audit

Males + cross
+ pride!

▲ CROSS — SHOWS AS

- X-linked: males show + criss-cross + verify. Ex: queen Victoria.
- List: hemophilia + colour blindness + Duchenne + G6PD + audit.

⚠ FATAL TRAP

Bleed-only + 'linked asked' + male-miss + cross-miss =
wrong + fail! Males + cross + audit.

★ NORCET HIT

HIT: Down + trisomies next — 21/18/13 + audit!

► Males + cross + carrier + list + sort?

X-linked + males + audit.

21/18/13 + audit! Flat/upslant + severe + audit — extra + signs + audit!

**Down 21 + signs
+ verify**

**Edwards 18 + Patau
13 + audit**

**Extra + signs
+ pride!**

◆ EXTRA — SIGNS AS

- Down: 21 + flat/upslant/single-crease + verify. Ex: screening + echo.
- Edwards 18 + Patau 13 + severe + audit. Ex: order of survival.

⚠ FATAL TRAP

Face-only + 'trisomy asked' + number-miss + sign-miss
= wrong + fail! Number + signs + audit.

★ NORCET HIT

HIT: Turner + Klinefelter next — XO/XXY + audit!

► Number + signs + 21/18/13 + extra + sort?

Trisomy + signs + audit.

XO + XXY + audit! Webbed/tall + hormones + audit — XO + XXY + audit!

Turner XO + webbed
+ short + verify

Klinefelter XXY +
tall + audit

XO + XXY
+ pride!

Σ XO — XXY AS

- Turner: XO + webbed + short + shield chest + verify.
- Klinefelter: XXY + tall + gynaecomastia + audit. Ex: infertility.

⚠ FATAL TRAP

Short-only + 'XO asked' + karyo-miss + sign-miss =
wrong + fail! Karyo + signs + audit.

★ NORCET HIT

HIT: Pedigree next — symbols + read + audit!

► Karyo + signs + XO/XXY + infertile + sort?

Turner + Kline + audit.

Symbols + read + audit! Square/circle + shade + audit — draw + read + audit!

**Square/circle +
shade + verify**

**Vertical/horizontal +
read + audit**

**Draw + read
+ pride!**

★ DRAW — READS AS

- Symbols: square male + circle female + shade affected + verify.
- Read: vertical AD + horizontal AR + males X-linked + audit.

⚠ FATAL TRAP

Tree-only + 'pedigree asked' + symbol-miss +
read-miss = wrong + fail! Symbols + read + audit.

★ NORCET HIT

HIT: Counseling next — guide + test + audit!

► Symbols + read + shade + pattern + check?

Pedigree + read + audit.

GENETIC COUNSELING + PRENATAL TESTS

Guide + test + audit! Amnio/CVS + ethics + audit — guide + test + audit!

Guide/support +
nondirective + verify

Amnio/CVS/USG +
tests + audit

Guide + test
+ pride!

▲ GUIDE — TESTS AS

- Counsel: nondirective + support + recurrence + verify. Ex: thalassemia pair.
- Tests: amnio/CVS/USG/NT + timing + audit. Ex: 11-14 NT + 15-20 amnio.

⚠ FATAL TRAP

Test-only + 'counsel asked' + guide-miss + timing-miss
= wrong + fail! Guide + timing + audit.

★ NORCET HIT

HIT: Therapy + revision next — fix + close + audit!

► Guide + timing + support + test + link?

Counsel + tests + audit.

GENE THERAPY + PHARMACOGENETICS

Fix + match + audit! Replace + response + nurse + audit — fix + match + audit!

Replace/add +
fix + verify

Drug-response +
match + audit

Fix + match
+ pride!

◆ FIX — MATCHES AS

- Therapy: replace/add + trials + verify. Ex: SMA + hemophilia hope.
- Pharmacogenetics: genes + drug response + audit. Ex: dose by genes.

⚠ FATAL TRAP

Gene-only + 'therapy asked' + fix-miss + match-miss =
wrong + fail! Fix + match + audit.

★ NORCET HIT

HIT: Revision tables next — patterns + close + audit!

► Fix + match + replace + dose + link?

Therapy + genes + audit.

Patterns + close + audit! AD/AR/X + numbers + audit — patterns + close + audit!

**AD/AR/X +
tables + verify**

**Numbers/disorders +
close + audit**

**Patterns + close
+ pride!**

Σ PATTERNS — CLOSE AS

- AD 50 + AR 25 + X males + tables + verify.
- Down 21 + Turner XO + Kline XXY + mocks + audit. Book closed.

⚠ FATAL TRAP

Read-only + 'close asked' + pattern-miss +
number-miss = wrong + fail! Patterns + numbers +
audit.

★ NORCET HIT

HIT: Book closed: verify + mock + genetics + audit!

► Patterns + numbers + tables + close + done?

Close + mock + audit.